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Profs. Enzo Aljovin, Hossein Movasati, Roberto Villafior Loyola

Dear Professors,

I am writing about your paper *Integral Hodge conjecture for Fermat varieties* (J. Symbolic Computation 95 (2019) 177–184; arXiv:1711.02628), with one verification and one small correction that may be useful for your records.

The verification. Using an independent instrument — the Gram matrix of the $3m^2$ lines built from the Schütt–Shioda–van Luijk intersection rules, with p -adic Smith reduction — I reproduced the surface rows of your Table 1 byte-exact (ranks and full elementary-divisor profiles), and extended the computation blind to degrees 16–20, where the results match a closed exponent law that your twelve rows satisfy: $|\text{disc } V| = m^{3(m-3)^2}$ for m odd (Shioda’s 1987 conjecture, verified to 81 by Schütt–Shioda–van Luijk) and $m^{3(m-3)^2} \cdot (m/4)^3$ for m even — the even half of which I could not find stated anywhere; a short note with sealed-before-computation blind tests (five cells, seven exponents, seven exact hits) is in my repository.

The correction. The row (6, 4) of your Table 2 prints

$$-1^{576} \cdot 2^{68} \cdot 4^{49} \cdot 8^{296} \cdot 16^{48} \cdot 32^{69} \cdot 512^2,$$

whose multiplicities sum to 1108. The rank of the lattice of primitive Hodge cycles of X_4^6 is 1107: it is the value of the closed-form rank of Degtyarev–Shimada (Remark 4.4 of arXiv:1405.4683), which I have measured independently in a separate computational campaign on that exact cell. As a consistency check, the same row-sum test passes on the other three rows of your Table 2 — (6, 3) sums to 70, (8, 3) to 252, (10, 3) to 924, each equal to the corresponding rank (the last one also measured independently in my campaign). So exactly one printed exponent in the (6, 4) row appears to be off by one (e.g. 1^{575} or 32^{68} would restore consistency); your raw data, referenced in §4 of the paper, would identify which. None of the theorems of the paper are affected — your Theorem 1 compares the machine-computed lists, not the typeset table.

Your Table 1 has been an unexpectedly fertile object: read jointly with the surface literature it contains the even half of the discriminant law above, the divisor profiles obey a law finer than the determinant one, and your data made both visible. Thank you for publishing it in full.

With best regards,

Rafael Amichis Luengo